Cobweb Forests:
A Kolmogorov–Arnold Theory of Networks of Concept Hierarchies

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Abstract

The Kolmogorov–Arnold (KA) representation theorem states that every continuous function of n variables is a finite superposition of continuous functions of one variable and addition, and Kolmogorov–Arnold Networks (KANs) turn that statement into an architecture with learnable univariate functions on its edges. We ask the two analogous questions for Cobweb, the incremental concept-formation system whose category utility is the mutual information between features and concepts. *Representation:* is there a set of Cobweb trees that represents an arbitrary continuous function? *Learnability:* does Cobweb's own learning procedure—category utility and its four operators, with no gradients—learn it? We formalise a *Cobweb forest*: a directed acyclic graph of hybrid Cobweb trees in which each tree reads raw features and the *messages* of other trees—their *activations* (multi-node predictions of an attribute) and their *labels* (the concepts an input sorts to)—and one designated tree produces the output. On representation we prove that a single tree is a universal approximator under two of the implementation's four prediction weightings (under the other two the natural construction keeps a fixed bias, and so do learned trees), a Cobweb–Kolmogorov–Arnold theorem (forests of two-input trees passing activations are universal with $4n^2 + 4n$ trees, and one input per tree is not enough), that label forests are universal, and that merging trees under a common root yields mixtures, not compositions. On learnability we prove that no Cobweb operator ever deletes a leaf, which makes a learned tree a universally consistent regressor whatever structure category utility builds, and we introduce *cognitive tasks*: what each tree perceives (its *cue*) and what it predicts and later observes (its *outcome*). A forest learned by Cobweb converges to the *ideal forest* of its task assignment, which equals the target exactly when the assignment is *sufficient*; predicting the target at every tree is sufficient for monotone-additive targets, predicting another view of the input is sufficient under multi-view redundancy, and categorization is always sufficient but at a rate governed by the dimension. Tasks cannot be inferred locally—a sign construction makes a tree's own data identical whether its input is necessary or irrelevant—but they can be inferred consistently by verifying candidate tasks at the tree that composes them. An executable verifier tests the statements against the implementation, and an independent reviewer, calibrated on a copy with planted errors, audits the proofs.

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1 Introduction

Kolmogorov–Arnold. For every $n \geq 2$ there are continuous functions $\varphi_{q,p} : [0,1] \to \mathbb{R}$ ($q = 0, \ldots, 2n$, $p = 1, \ldots, n$) such that every $f \in C([0,1]^n)$ can be written as

$$f(x_1, \ldots, x_n) = \sum_{q=0}^{2n} \Phi_q\Big(\sum_{p=1}^{n} \varphi_{q,p}(x_p)\Big) \tag{1}$$

with continuous outer functions $\Phi_q : \mathbb{R} \to \mathbb{R}$ that depend on f [1, 12, 14, 21]. The only genuinely multivariate operation is addition. The inner functions may be very rough (nowhere differentiable), which long made (1) look irrelevant to learning [9]. Kolmogorov–Arnold Networks [13] stack layers of learnable univariate functions (B-splines on edges, sums at nodes) and prove an approximation theorem whose rate does not depend on n when f has a *smooth* KA-type representation [13, Thm. 2.1].

Cobweb. Cobweb [8] sorts each instance down a hierarchy of probabilistic concepts and, at every node, applies whichever of four operators—add to the best child, create a new child, merge the two best children, split the best child—most increases category utility, which in its information-theoretic form is the mutual information between features and concepts [4, 17]. Concepts store sufficient statistics (attribute–value counts for nominal attributes, a mean and variance for each continuous one), so learning is closed-form, incremental and does not forget. Prediction expands concepts best-first and mixes their predictions [17].

Several recent systems use Cobweb trees as *modules* whose representations become the inputs of other trees: Convolutional Cobweb [16] sorts image patches into a filter hierarchy and describes each image by the patches' filter labels for a second, classification hierarchy; TRELLIS [20] describes composites by bags of concept identifiers drawn from another hierarchy; and a Cobweb language model passes each tree's prediction for a position to the trees above it. We call such a network a *Cobweb forest* and ask what it can represent and learn.

Questions.

Q1. *Representation.* For every continuous f and $\varepsilon > 0$, is there a Cobweb forest whose output is uniformly within ε of f? How few trees, and how few inputs per tree, suffice?

Q2. *Learnability.* If every tree learns only with Cobweb's own procedure, does the forest's output converge to f? What must each tree be asked to predict, and can that be inferred from data?

Representation is not learnability. The constructions that answer Q1 choose what every tree holds knowing f: the trees of the KA construction hold the values $\varphi_{q,p}(x_p)$ and $\Phi_q(u)$. A learner never observes these values; it observes its inputs and, eventually, outcomes. A KAN recovers the individual terms of a sum from the error of the sum by back-propagation. A Cobweb tree has no gradient: it can only learn to predict what it will observe. Q2 is therefore a question about *tasks*—what each tree perceives and what it predicts—and the central objects of the second half of the paper are *cognitive tasks* (Section 5) and their inference (Section 6).

Contributions.

- A formal model of the hybrid Cobweb tree as implemented in `cobweb_hybrid`, including its four prediction weightings, and of Cobweb forests with activation and label messages (Section 2). An executable verifier checks that the model equals the code (Appendix A).
- Representation (Section 3): single-tree universality with an explicit rate (Theorem 9); inconsistency of the prior-weighted all-concept mixture (Proposition 10); the Cobweb–Kolmogorov–Arnold theorem (Theorem 12) and the necessity of fan-in two (Proposition 13); universality of label forests (Theorem 15); merging trees gives mixtures (Proposition 16).
- One learned tree (Section 4): no operator deletes a leaf (Lemma 17), so a learned tree is universally consistent whatever structure category utility builds (Theorem 18).
- Cognitive tasks (Section 5): tasks as cue–outcome pairs with observable outcomes (Definition 20); a learned forest converges to the ideal forest of its tasks (Theorem 24) and is consistent if and only if the tasks are sufficient (Corollary 25); funnel tasks are sufficient for monotone-additive targets (Proposition 26) but not for x_1x_2 (Proposition 27); cross-view tasks are sufficient under multi-view redundancy (Proposition 28); categorization is always sufficient, at a dimension-dependent rate (Theorem 29).

- Inferring tasks (Section 6): no rule that looks at a tree's own data can infer its task (Theorem 34, Corollary 35); verifying candidate tasks at the composing tree selects a sufficient assignment with probability tending to one (Theorem 36); this yields task operators chosen like Cobweb's own (Procedure 38).

2 The model

2.1 Concepts and trees

An *instance* is a pair $z = (a, u)$ of a nominal part a (a table that assigns to each attribute a bag of values with weights) and a continuous part $u \in \mathbb{R}^m$.

Definition 1 (Concept)**.** A concept c stores its count n_c, the attribute–value counts $n_c(A{=}v)$ of the instances it covers, and, for the continuous block, their mean $\mu_c \in \mathbb{R}^m$ and empirical (per-dimension) variance $s_c^2 \in \mathbb{R}^m_{\geq 0}$. With smoothing parameters $\alpha > 0$ and $h^2 > 0$ (`alpha`, `prior_var`) a concept scores a query with

$$P_c(A{=}v) = \frac{n_c(A{=}v) + \alpha}{n_c(A) + (K_A + 1)\alpha}, \qquad p_c(u_j) = \mathcal{N}\big(u_j;\, \mu_{c,j},\, v_{c,j}\big), \quad v_{c,j} = s_{c,j}^2 + h^2,$$

with attributes conditionally independent given c and K_A the number of values of A seen so far (the extra α reserves mass for an unseen value). When a concept *predicts* a missing attribute it uses $(n_c(A{=}v) + \alpha)/(n_c(A) + K_A\alpha)$, which sums to one over the seen values.

Definition 2 (Cobweb tree)**.** A Cobweb tree $\mathcal{T}$ is a rooted tree of concepts whose statistics are *additive*: each internal concept's statistics are those of the pooled instances of its children. Its *category utility* at a node with children C is the normalised information gain $\mathrm{PU} = |C|^{-1}\big[H(\text{parent}) - \sum_{c \in C} P(c)H(c)\big]$, where H is the sum of the Dirichlet-smoothed categorical entropies and of the Gaussian differential entropies $\frac{1}{2}\log(2\pi e\, v_{c,j})$. For nominal attributes and $\alpha \to 0$ the bracket is $\sum_A \big[H(A) - H(A \mid C)\big] = \sum_A I(A; C)$, the information the partition carries about each attribute; the Gaussian terms are its analogue for fitted Gaussians. This is the information-theoretic category utility of Corter and Gluck [4], which the rational account reads as the mutual information between features and concepts [17].

Learning (`ifit`) sorts an instance from the root. At a leaf it either increments the leaf (exact match) or performs a *fringe split*: the leaf becomes an internal node with the old leaf and a new leaf for the instance as children. At an internal node it adds the instance and chooses, by category utility, between BEST (descend into the best child), NEW (create a leaf), MERGE (join the two best children under a new node) and SPLIT (replace the best child by its children).

2.2 Prediction

A *query* is an instance in which some attributes and some continuous coordinates are missing; we write O for the observed coordinates (missing continuous coordinates are encoded as `NaN`, missing attributes are absent). The log-likelihood of a query under a concept is $\ell_c(q) = \sum_{A\ \text{obs.}} \log P_c(A{=}q_A) + \sum_{j \in O} \log p_c(q_j)$.

Definition 3 (Multi-node prediction)**.** Prediction expands concepts best-first from the root, scoring each non-root concept c with one of

$$\begin{aligned}
&\text{LIK}: && s_c = \ell_c(q),\\
&\text{PRIOR}: && s_c = \ell_c(q) + \log(n_c/n_{\text{root}}),\\
&\text{PATH}: && s_c = \pi_c(q) + \ell_c(q), \qquad \pi_c(q) = \textstyle\sum_{c' \in \text{path}(c)\setminus\{\text{root}\}} \log \frac{n_{c'} e^{\ell_{c'}(q)}}{\sum_{b \in \text{sib}(c')} n_b e^{\ell_b(q)}},\\
&\text{POSTERIOR}: && s_c = \pi_c(q),
\end{aligned}$$

where $\text{sib}(c')$ are the children of c''s parent (including c'); under every weighting the root is scored by its likelihood, $s_{\text{root}} = \ell_{\text{root}}(q)$. With $E_M(q)$ the expanded set and $w_c(q) \propto e^{s_c}$ normalised over $E_M(q)$, the prediction of a missing continuous coordinate j and of a missing attribute A is

$$\hat{y}_j(q) = \sum_{c \in E_M(q)} w_c(q)\, \mu_{c,j}, \qquad \hat{P}(A{=}v \mid q) = \sum_{c \in E_M(q)} w_c(q)\, P_c(A{=}v).$$

The budget M bounds the number of expanded concepts. The *leaves-only* variant restricts the mixture to the expanded leaves; its budget counts leaves (expansion stops once M leaves have entered the mixture, after at most $64M$ expansions). *Full expansion* is $M = \infty$.

PATH is the default of the implementation and of `cobweb_torch`; $\pi_c(q)$ is the log posterior of c obtained by multiplying the sibling-normalised posteriors down its path, so $\sum_{c \text{ at depth } k} e^{\pi_c} \le 1$ for every depth k. The rational account [17] ranks concepts by pointwise mutual information $\log p(x \mid c) - \log p(x)$, i.e. by ℓ_c.

2.3 Cobweb forests

Definition 4 (Messages)**.** Let $\mathcal{T}$ be a tree whose instances contain *outcome* attributes $V_{\mathcal{T}}$ (possibly none; Definition 20). For a query q in which $V_{\mathcal{T}}$ is missing:

- the *activation* of $\mathcal{T}$ is its prediction of $V_{\mathcal{T}}$—the vector $\hat{y}_{V_{\mathcal{T}}}(q)$ for a continuous outcome, the predicted distribution $(\hat{P}(V_{\mathcal{T}}{=}v \mid q))_v$ for a nominal one;
- the *label* of $\mathcal{T}$ at depth k is the identifier of the depth-k concept on the root-to-leaf descent that chooses, at each node, the child maximising $\log n_c + \ell_c(q)$ (the leaf if the path is shorter; $k = \infty$ denotes the leaf).

Definition 5 (Cobweb forest)**.** A Cobweb forest $\mathfrak{F} = (V, E, o)$ is a finite directed acyclic graph whose vertices are Cobweb trees and whose edges $\sigma \to \tau$ carry the activation and/or the label of σ into τ, together with an output tree $o \in V$. A tree τ learns instances made of (a) the raw coordinates x_{S_τ} it is wired to, (b) the activations of its parents as continuous coordinates, (c) the labels of its parents as nominal attributes (one attribute per parent), and (d) its outcome, if any. For an input x the forest computes the messages of every tree in topological order and outputs the activation of o. The *fan-in* of τ is the number of its non-outcome inputs (raw coordinates, activation coordinates, label attributes).

Definition 6 (Training protocols)**.** Every tree learns only with `ifit`. Two protocols fix the messages a tree is trained on:

- *online*: for each observation i, every tree first emits its message for x_i from what it has learned from observations $1, \dots, i-1$, then learns observation i (messages are out-of-sample, as at test time);

- *split*: the stream is cut into consecutive chunks, one per level of the DAG; a tree at level k learns chunk k, reading the messages of its already-trained (frozen) parents.

Remark 7 (Why not train stage-wise on the same data). A tree that has learned (x_i, y_i) keeps a leaf for it (Lemma 17), and its activation at x_i is close to y_i itself. A parent trained on such in-sample messages learns the relation between y and a leaked copy of y, which it never sees at test time. The online protocol (predict, then learn), already used by the Cobweb language model, and the split protocol avoid this.

3 Representation

Throughout, a tree that *represents* $g : K \to \mathbb{R}$ ($K \subset \mathbb{R}^d$ compact) is a tree over instances $(x, g(x))$ queried with the last coordinate missing, and $\omega_g(r) = \sup\{|g(x) - g(x')| : \|x - x'\| \le r\}$ is the modulus of continuity.

3.1 A local error certificate

Lemma 8 (Concept bias). *Let c be a concept of a tree over instances $(x_i, g(x_i))$ with g L-Lipschitz. Then $|\mu_{c,Y} - g(\mu_{c,X})| \le L\sqrt{\operatorname{tr} S_c}$, where S_c is the empirical covariance of the x-part of c's instances, and consequently, for every query x, every weighting and every budget,*

$$|\hat{y}(x) - g(x)| \;\le\; L \sum_{c \in E_M(x)} w_c(x) \Big(\|x - \mu_{c,X}\| + \sqrt{\operatorname{tr} S_c} \Big). \tag{2}$$

Proof. $|\mu_{c,Y} - g(\mu_{c,X})| = |\frac{1}{n_c}\sum_i (g(x_i) - g(\mu_{c,X}))| \le \frac{L}{n_c}\sum_i \|x_i - \mu_{c,X}\| \le L(\frac{1}{n_c}\sum_i \|x_i - \mu_{c,X}\|^2)^{1/2}$ by Jensen. For (2), write $\hat{y} - g(x) = \sum_c w_c(\mu_{c,Y} - g(\mu_{c,X})) + \sum_c w_c(g(\mu_{c,X}) - g(x))$ and use $\sum_c w_c = 1$. □

The right side of (2) is computable from the tree: a mixture is accurate when it puts its weight on concepts that are near the query *and* tight.

3.2 One tree

Theorem 9 (A single tree is a universal approximator). *Let $g \in C(K)$, $K \subset \mathbb{R}^d$ compact, and $\varepsilon > 0$. There is a Cobweb tree $\mathcal{T}$ and a variance floor $h^2 > 0$ such that $\sup_{x \in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| < \varepsilon$ under full expansion with each of the weightings* LIK *and* PATH*, and with the leaves-only variant of any weighting. If g is L-Lipschitz, a tree with G leaves achieves*

$$\sup_{x \in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| \;\le\; C_d\, L\, G^{-1/d} \sqrt{\log G} + C_d\, \|g\|_\infty \big(G^{-2/d} + G^{-1} \big). \tag{3}$$

Proof. Extend g continuously to a cube $Q \supseteq K$ (Tietze) and rescale $Q = [0,1]^d$. Fix $\beta > 0$, let $\delta = \beta h$ and let Γ be the grid of cell centres of spacing δ, $G = |\Gamma| = \delta^{-d}$. Take the depth-one tree whose leaves are the instances $(\gamma, g(\gamma))$, $\gamma \in \Gamma$, and whose root pools them. Each leaf has $s^2 = 0$, hence variance h^2 in every coordinate; the root has variance at least $v_0 > 0$ (the variance of the grid) in every coordinate. Write $k_\gamma = (2\pi h^2)^{-d/2} e^{-\|x-\gamma\|^2/2h^2}$.

Leaves dominate. Some γ lies within $\sqrt{d}\,\delta/2$ of x, so $\max_\gamma k_\gamma \ge (2\pi h^2)^{-d/2} e^{-d\beta^2/8}$, while $\sum_\gamma k_\gamma \le (2\pi h^2)^{-d/2} C(\beta, d)$ by comparison with a Gaussian sum over a grid of spacing β in units of h. Under LIK the leaves carry unnormalised mass $\sum_\gamma k_\gamma \ge c\, h^{-d}$ and the root at most $(2\pi v_0)^{-d/2}$. Under PATH the leaf γ has $e^{\pi_\gamma} = k_\gamma / \sum_{\gamma'} k_{\gamma'}$ (equal counts), so its mass is $k_\gamma^2 / \sum k$, whose sum is

at least $(\max k)^2/\sum k \geq c'h^{-d}$, while the root again has mass at most $(2\pi v_0)^{-d/2}$. In both cases the root's weight is $O(h^d) = O(G^{-1})$.

Leaves form a kernel smoother. Among the leaves, $w_\gamma \propto k_\gamma^r$ with $r = 1$ (LIK) or $r = 2$ (PATH). For any $\rho > 0$, splitting the exponent and using the lower bound on $\max_\gamma k_\gamma^r$ for the normaliser,

$$\sum_{\|\gamma - x\| > \rho} w_\gamma \;\leq\; \frac{e^{-r\rho^2/4h^2} \sum_\gamma e^{-r\|x-\gamma\|^2/4h^2}}{e^{-rd\beta^2/8}} \;\leq\; C'' e^{-r\rho^2/4h^2},$$

so $|\sum_\gamma w_\gamma g(\gamma) - g(x)| \leq \omega_g(\rho) + 2\|g\|_\infty C'' e^{-\rho^2/4h^2}$. Taking $\rho = \sqrt{h}$ makes this vanish as $h \to 0$, uniformly in x; adding the root's $O(h^d)$ weight (times $2\|g\|_\infty$) gives the first claim. For Lipschitz g take $\rho = 2h\sqrt{2\log(1/h)}$, so that $\omega_g(\rho) \leq L\rho$ and the tail is at most $C''h^2$; with $h \asymp G^{-1/d}$ this is (3). The leaves-only variant drops the root term; restricted to the leaves of this tree, PRIOR coincides with LIK (equal counts) and POSTERIOR weights the leaves by $k_\gamma/\sum k$, i.e. $r = 1$. □

The PRIOR weighting behaves differently: it multiplies each concept's likelihood by its *relative count*, and the internal concepts then keep a share of the mixture that does not vanish as the variance floor shrinks.

Proposition 10 (The prior- and posterior-weighted all-concept mixtures are inconsistent)**.** *In the setting of the proof of Theorem 9, with* PRIOR *weighting and full expansion, the root's weight is* $\lambda(x) = \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + \theta_h(x))$*, where* $\mathcal{N}_{root}(x) = \mathcal{N}(x; \mu_{root}, v_{root})$ *and* $\theta_h(x) = G^{-1}\sum_\gamma k_\gamma$ *is bounded above by a constant* $C(\beta, d)$ *independent of* h*. Hence* $\lambda(x) \geq \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + C(\beta, d))$ *for all* h*, and* $\hat{y}(x) - g(x) = \lambda(x)\big(\mu_{root,Y} - g(x)\big) + o(1)$*: the error does not vanish wherever* $g(x)$ *differs from the grid mean of* g*. With* POSTERIOR *weighting the leaves' masses* e^{π_γ} *sum to exactly one, so* $\lambda(x) = \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + 1)$ *and the same conclusion holds.*

Proof. Leaf γ has prior $1/G$, so the leaves' mass is $\theta_h(x) = G^{-1}\sum_\gamma k_\gamma = \beta^d(2\pi)^{-d/2}\sum_\gamma e^{-\|x-\gamma\|^2/2h^2}$ (using $G^{-1} = \delta^d$), a Riemann sum of mesh β of the standard Gaussian density, bounded by $C(\beta, d)$ and bounded below by a positive constant (it is close to 1 in the interior when β is small, but it does not converge as $h \to 0$; it oscillates with x/h). The root has prior 1 and mass $\mathcal{N}_{\text{root}}(x)$, which tends to a positive limit. The leaves' mixture converges to $g(x)$ as in Theorem 9. Under POSTERIOR the leaves are the root's children and their path posteriors are the sibling-normalised posteriors, which sum to one, while the root is scored by its likelihood (Definition 3). □

In a learned tree the defect of POSTERIOR is compounded: every non-root concept is scored by its path posterior $\pi_c \leq 0$, which can only decrease down a path, so the wide concepts near the root outweigh the leaves whatever h is. Both are *all-concept* pathologies; restricted to leaves the weightings converge in the construction above, and Section 7 measures them in learned trees.

3.3 Composition: the Cobweb–Kolmogorov–Arnold theorem

Lemma 11 (Approximating a DAG of continuous maps)**.** *Let* D *be a finite DAG of continuous maps* $g_v : \mathbb{R}^{k_v} \to \mathbb{R}$ *whose inputs are raw coordinates of* $x \in [0,1]^n$ *or outputs of other vertices, and let* F *be the map it computes; let* K_v *be the (compact) set of input vectors of* v *as* x *ranges over* $[0,1]^n$*. For every* $\varepsilon > 0$ *there are* $\rho_v, \eta_v > 0$ *such that any maps* $\hat{g}_v$ *with* $\sup_{K_v + \bar{B}_{\rho_v}} |\hat{g}_v - g_v| \leq \eta_v$ *define a network* $\hat{F}$ *with* $\sup_x |\hat{F}(x) - F(x)| \leq \varepsilon$*.*

Proof. Choose the tolerances in reverse topological order. At the output vertex put $\eta_o = \varepsilon/2$ and choose $\rho_o \leq 1$ with $\omega_{g_o}(\rho_o) \leq \varepsilon/2$ on the compact $K_o + \bar{B}_1$. Given the tolerance $\rho_v \leq 1$ by which the

input vector of v may deviate, allow each of its k_v inputs to deviate by $\rho_v/\sqrt{k_v}$: for each incoming vertex u set $\eta_u = \rho_v/(2\sqrt{k_v})$ and choose $\rho_u \le 1$ with $\omega_{g_u}(\rho_u) \le \rho_v/(2\sqrt{k_v})$ on $K_u + \bar{B}_1$, taking minima when u feeds several vertices. Then, in topological order, every vertex's input deviates from its exact value by at most ρ_v, so its output deviates by at most $\eta_v + \omega_{g_v}(\rho_v)$, which is within its consumers' tolerance; at the output the deviation is at most ε. □

Theorem 12 (Cobweb–Kolmogorov–Arnold)**.** *Let $n \ge 2$. For every $f \in C([0,1]^n)$ and $\varepsilon > 0$ there is a Cobweb forest with activation messages, in which every tree has at most two inputs, consisting of*

- *$n(2n+1)$ univariate trees for the inner functions $\varphi_{q,p}$ (independent of f),*
- *$(2n+1)(n-1)$ two-input trees forming the inner sums,*
- *$2n+1$ univariate trees for the outer functions Φ_q, and*
- *$2n$ two-input trees forming the outer sum,*

$4n^2 + 4n$ trees in all, whose output is within ε of f on $[0,1]^n$.

Proof. Write f in the form (1) and read it as a DAG of continuous maps of one or two variables: $\varphi_{q,p}$, the binary additions $(a,b) \mapsto a+b$ chained to form the $2n+1$ inner sums of n terms and the outer sum of $2n+1$ terms, and Φ_q. Apply Lemma 11, and realise each $\hat{g}_v$ on the compact $K_v + \bar{B}_{\rho_v}$ by Theorem 9 as a tree over instances $(\text{inputs}, g_v(\text{inputs}))$; its activation is the required $\hat{g}_v$. The count is $n(2n+1) + (2n+1)(n-1) + (2n+1) + 2n = 4n^2 + 4n$. □

Proposition 13 (Two inputs are necessary)**.** *If every tree of a forest has fan-in at most one, its output depends on at most one raw coordinate. Hence fan-in two is the least fan-in at which Cobweb forests are universal.*

Proof. By induction in topological order, each tree's messages are functions of the at most one raw coordinate at the source of its unique input chain. □

Remark 14 (Rates and smooth KA representations). Theorem 12 uses a fixed number of trees, but for an arbitrary f the inner functions of (1) are rough, and the number of leaves each tree needs is not controlled. As for KANs [13, Thm. 2.1], the useful statement assumes structure. If f is computed by a DAG of s maps of fan-in $\le k$, each L-Lipschitz on a neighbourhood of its input set, then (3) and the stability argument of Lemma 11 give a forest of s trees with G leaves each and error $C(s,L)\, G^{-1/k}\sqrt{\log G}$—a rate set by the fan-in k, not by n. A single tree over the n inputs has rate $G^{-1/n}$ (Theorem 9). Cobweb trees realise sums as learned two-input maps; KANs sum at nodes for free, which is why their rate is governed by one-dimensional splines.

3.4 Label forests

Theorem 15 (Label forests are universal)**.** *For every $f \in C([0,1]^n)$ and $\varepsilon > 0$ there is a forest of n univariate trees without outcomes that pass leaf labels to an output tree whose instances consist of the n labels (nominal attributes) and the target, such that the output tree's prediction (LIK, leaves only, full expansion) is uniformly within ε of f.*

Proof. Let tree p be the depth-one tree over the K grid points of spacing $\delta = 1/K$ in $[0,1]$, all with count one and variance h^2. Its label for x_p is $\arg\max_\gamma\{\log n_\gamma + \ell_\gamma(x_p)\}$, the nearest grid point, so the labels partition $[0,1]^n$ into boxes B_t of side δ, indexed by label tuples t. Let the output tree be the depth-one tree with one leaf per tuple t, holding the instance $(t, f(\text{centre}(B_t)))$. For a query with tuple t, a leaf t' that disagrees with t in m attributes has likelihood ratio $\big(\alpha/(1+\alpha)\big)^m$ relative to leaf

t (Definition 1, counts one). There are at most $\binom{n}{m}(K-1)^m$ such leaves, so their total weight is at most $\sum_{m\geq 1}\binom{n}{m}(K\alpha)^m = (1+K\alpha)^n - 1$. Hence $|\hat{y}(x) - f(x)| \leq \omega_f(\sqrt{n}\,\delta/2) + 2\|f\|_\infty\big((1+K\alpha)^n - 1\big)$, which is below ε for K large and then α small. □

The label forest of Theorem 15 has an output tree with K^n leaves: labels carry no metric, so this construction pays for every cell of the product partition. Activations carry the metric and allow the fan-in-two construction of Theorem 12.

3.5 Interweaving trees

Proposition 16 (Merging trees under a common root gives mixtures)**.** *Let $\mathcal{T}_1, \ldots, \mathcal{T}_m$ be trees over the same instance space with the same value sets (e.g. continuous-only trees), and let $\mathcal{T}$ be the tree whose root pools their roots' statistics and has $\mathcal{T}_1, \ldots, \mathcal{T}_m$ as its subtrees. Under full expansion with* LIK *or* PATH *weighting,*

$$\hat{y}_{\mathcal{T}}(q) = \lambda_0(q)\,\mu_{root} + \sum_{j=1}^{m} \lambda_j(q)\,\hat{y}_{\mathcal{T}_j}(q), \qquad \lambda_j \geq 0,\ \sum_{j=0}^{m} \lambda_j = 1,$$

with $\lambda_0 \propto e^{\ell_{root}}$ and $\lambda_j \propto Z_j$ (LIK) or $\lambda_j \propto P(root_j \mid q)\,Z_j$ (PATH), where Z_j is the total unnormalised weight of $\mathcal{T}_j$'s own mixture and $P(root_j \mid q)$ the sibling-normalised posterior of $\mathcal{T}_j$'s root.

Proof. Under LIK a concept's score does not depend on the tree it sits in, so the mixture over $\mathcal{T}$ splits into the root term and the m subtree mixtures. Under PATH every concept of $\mathcal{T}_j$ gains the same term $\log P(\text{root}_j \mid q)$ in π_c (the sibling sets below root_j are unchanged), which rescales the subtree's weights uniformly and leaves its internal mixture unchanged. With different value sets the smoothing denominators K_A change, and the identity holds only approximately. □

Interweaving therefore closes the class of tree predictors under input-dependent convex combinations; it never composes them. Composition—one tree reading another's representation—is what forests add.

4 Learnability of one tree

Lemma 17 (Leaf preservation)**.** *Run* `ifit` *on any sequence $z_1, \ldots, z_N$ with any choices of operators (including add-only training), and suppose the depth limit is never reached. Then (i) every concept's statistics are the pooled statistics of the instances stored in the leaves below it; (ii) no leaf is ever deleted; (iii) every instance is stored in exactly one leaf, and the instances stored in a leaf are identical up to the exact-match tolerance. In particular, if the instances are pairwise distinct, the leaves are in bijection with them.*

Proof. By induction over the branches of the sorting loop. BEST increments the current node and descends; NEW increments the current node and creates a leaf holding z; MERGE inserts a new internal node whose statistics are the pooled statistics of the two children it adopts; SPLIT deletes the best child, which is internal (the operator requires it to have at least two children), and promotes its children; at a leaf, an exact match increments it and any other instance triggers a fringe split, which copies the leaf's statistics into a new internal parent, keeps the old leaf, and adds a leaf for z. No branch deletes a leaf, every instance ends in exactly one leaf, and a leaf only absorbs exact matches. Two copies of the same instance need not share a leaf: sorting may lead the second copy to a different leaf, which it then splits. □

When the depth limit is reached the implementation inserts the instance into the current node, which may be a leaf; (iii) then fails, which is why the lemma assumes the default limit (1000) is not reached.

Theorem 18 (A learned tree is universally consistent)**.** *Let (X_i, Y_i) be i.i.d. with $X \in \mathbb{R}^d$, $|Y| \le B$, $m(x) = \mathbb{E}[Y \mid X = x]$. Let $\mathcal{T}_N$ be the tree that Cobweb learns from $(X_i, Y_i)_{i \le N}$ in any order and with any operator choices, and let it predict Y from X with the leaves-only mixture,* LIK *weighting and full expansion, with variance floor h_N^2. If X has a density, $h_N \to 0$ and $N h_N^d \to \infty$, then $\int |\hat{y}_{\mathcal{T}_N}(x) - m(x)|\, \mu(dx) \to 0$ almost surely, whatever structure category utility builds (μ the distribution of X).*

Proof. With a density the X_i are distinct almost surely and so are the instances, so by Lemma 17 the tree has exactly N leaves, leaf i holding (X_i, Y_i) with zero empirical variance. The LIK score of leaf i for a query x with Y missing is $\log \mathcal{N}(x; X_i, h_N^2 I)$, so the leaves-only mixture is $\sum_i K_{h_N}(x - X_i) Y_i / \sum_i K_{h_N}(x - X_i)$, the Nadaraya–Watson estimate with a Gaussian kernel. The Gaussian kernel is *regular* (it is bounded below by a positive constant on a ball and $\int \sup_{\|u-x\| \le r} K(u)\, dx < \infty$), and for regular kernels and bounded Y this estimate is strongly consistent in $L_1(\mu)$ under $h_N \to 0$, $N h_N^d \to \infty$ [7]; see Györfi et al. [11, Ch. 5] for the boxed-kernel version in L_2. □

Remark 19 (What the structure is for). Theorem 18 holds because Cobweb never forgets an instance; the hierarchy plays no role in it. The structure matters for three things the theorem does not cover: (a) a *budget* $M < \infty$, where best-first search must find the leaves near the query; (b) the PATH weighting, where the mixture weight of a leaf is its path posterior times its likelihood, so the leaves-only mixture is not a kernel smoother in a learned tree (Lemma 8 still bounds its error); and (c) the *labels* a tree passes to other trees, which are its concepts. The experiments use the implementation's default, PATH with a leaf budget; they are therefore a test of these structural effects as much as of the theorem. The branch-and-bound stop of `categorize(use_best=True)` assumes $\ell_c \le 0$, which holds for nominal attributes but not for Gaussian densities; with a continuous block it can return a concept whose score is below the best (Appendix A exhibits one).

5 Cognitive tasks

A Cobweb tree is unsupervised: it models the joint distribution of whatever its instances contain and predicts any attribute that is missing from a query. What a tree *passes on* is therefore fixed by what its instances contain. We make this the central object.

5.1 Tasks as cues and outcomes

Definition 20 (Cognitive task)**.** Let the stream consist of i.i.d. observations (X, Y), $X \in \mathbb{R}^n$ the raw input and Y the target. A *cognitive task* for a tree τ is a pair (C_τ, V_τ):

- the *cue* C_τ, what the tree perceives: its raw coordinates X_{S_τ} and the messages of its parents;
- the *outcome* V_τ, what the tree predicts and later observes: either nothing ($V_\tau = \varnothing$) or an *observable* variable, i.e. a coordinate of the observation that the cue does not contain—the target Y or a raw coordinate X_j, $j \notin S_\tau$.

The tree learns the instances (C_τ, V_τ) with `ifit` and is queried with V_τ missing (Definition 4). A *task assignment* A gives every tree of a forest a task; the output tree's outcome is Y.

Three families of tasks occur in the systems that motivate this paper. *Categorization* ($V_\tau = \varnothing$): the tree passes only labels, as the filter hierarchy of Convolutional Cobweb does. *Funnel* ($V_\tau = Y$): every tree predicts the final target, as every tree of the Cobweb language model predicts the next word. *Cross-view* ($V_\tau = X_j$, $j \notin S_\tau$): the tree predicts a part of the input it does not perceive—a neighbouring pixel, the next token, the context of a chunk, as TRELLIS's content hierarchy models its context. The cue–outcome reading is the one of associative learning, in which an organism learns to predict an outcome it will observe from the cues it perceives [19]; the online protocol of Definition 6 is literally "predict, then observe, then learn".

Observability is the constraint that separates learnable from representable forests.

Remark 21 (The KA tasks are not observable). In Theorem 12 the tree for Φ_q holds the values $\Phi_q(u_q)$. The stream never shows them: it shows only their sum $Y = \sum_q \Phi_q(u_q)$. A KAN separates the terms of the sum by back-propagating the error of the sum through each of them; a Cobweb tree can only predict what it will observe. The inner functions $\varphi_{q,p}$ do not depend on f, so a forest could be *given* them as fixed tasks, but the outer decomposition is not identifiable from the stream without further structure. The rest of this section asks which *observable* tasks recover f anyway.

5.2 What a learned forest converges to

Definition 22 (Ideal forest, sufficiency, deficit)**.** Let A be a task assignment of an activation forest. Define, in topological order, the *ideal cue* and *ideal message* of each tree,

$$U_\tau = \big(X_{S_\tau}, (m_\sigma(X))_{\sigma\to\tau}\big), \qquad m_\tau(X) = \mathbb{E}[V_\tau \mid U_\tau] = \psi_\tau(U_\tau),$$

and the *ideal output* $F_A(X) = m_o(X) = \mathbb{E}[Y \mid U_o]$. Let $f(X) = \mathbb{E}[Y \mid X]$. The assignment is *sufficient* if $F_A = f$ almost surely, and its *deficit* is

$$\Delta(A) = \mathbb{E}\big[(f(X) - F_A(X))^2\big] = \mathbb{E}\,\mathrm{Var}(Y \mid U_o) - \mathbb{E}\,\mathrm{Var}(Y \mid X).$$

The identity holds because U_o is a function of X, so $F_A = \mathbb{E}[f(X) \mid U_o]$ and $\mathrm{Var}(Y \mid U_o) = \mathbb{E}[\mathrm{Var}(Y \mid X) \mid U_o] + \mathrm{Var}(f(X) \mid U_o)$.

Assumption 23 (Regularity)**.** (A1) The observations are i.i.d., X has compact support, every outcome is bounded, $|V_\tau| \le B$, and the instances of every tree are almost surely pairwise distinct (e.g. V_τ has a continuous distribution). (A2) For every tree, the ideal cue $U_\tau \in \mathbb{R}^{k_\tau}$ has a density bounded away from zero and infinity on its support D_τ, a compact set with Lipschitz boundary, and ψ_τ extends to a uniformly continuous function on a neighbourhood of D_τ. (A3) Every tree predicts with the leaves-only LIK mixture under full expansion, with variance floor $h_{\tau,N}^2$, $h_{\tau,N} \to 0$ and $N h_{\tau,N}^{k_\tau} / \log N \to \infty$. (A4) For every edge $\sigma \to \tau$, $\varepsilon_{\sigma,N}/h_{\tau,N} \to 0$ in probability, where $\varepsilon_{\sigma,N} = \sup_x |\hat m_\sigma(x) - m_\sigma(x)|$ is the parent's uniform error.

(A4) is a condition on the schedule: by induction the parents' errors tend to zero (Theorem 24), so variance floors that shrink slowly enough satisfy it.

Theorem 24 (A Cobweb-learned forest converges to its ideal forest)**.** *Let $\mathfrak{F}$ be an activation forest with task assignment A, trained with the split protocol on N observations, and let Assumption 23 hold. Then for every tree τ, $\sup_x |\hat m_\tau(x) - m_\tau(x)| \to 0$ in probability; in particular the output satisfies $\sup_x |\hat F_N(x) - F_A(x)| \to 0$ in probability.*

Proof. Induction over the levels of the DAG. Fix a tree τ at level ℓ and condition on the chunks of the earlier levels, on which its parents were trained and frozen. Write $\hat u_\tau(x) = (x_{S_\tau}, (\hat m_\sigma(x))_\sigma)$ for its learned cue map and $u_\tau(x)$ for the ideal one, so $\|\hat u_\tau - u_\tau\|_\infty \le \varepsilon := \sqrt{k_\tau}\max_\sigma \varepsilon_{\sigma,N}$.

Step 1: the tree is a kernel smoother. The tree learns the instances $(\hat{U}_i, V_i) = (\hat{u}_\tau(X_i), V_{\tau,i})$ of chunk ℓ, which are i.i.d. given the earlier chunks and pairwise distinct by (A1). By Lemma 17 its leaves are these instances with zero spread, and its leaves-only LIK activation at a cue u is the Nadaraya–Watson estimate $\hat{\psi}(u) = \sum_i K_h(u - \hat{U}_i)V_i / \sum_i K_h(u - \hat{U}_i)$ with the Gaussian kernel of bandwidth $h = h_{\tau,N}$.

Step 2: the perturbed design does not matter. Let $\check{\psi}$ be the same estimate built on the ideal cues $U_i = u_\tau(X_i)$. By (A1)–(A3), $\check{\psi} \to \psi_\tau$ uniformly on the δ-neighbourhood of D_τ for any $\delta = o(h)$, almost surely [6, 15]. Write $\hat{U}_i = U_i + \Delta_i$, $\|\Delta_i\| \le \varepsilon$. For $\|u - U_i\| \le Rh$,

$$\frac{K_h(u - \hat{U}_i)}{K_h(u - U_i)} = \exp\Big(\frac{2\langle u - U_i, \Delta_i\rangle - \|\Delta_i\|^2}{2h^2}\Big) \in \big[e^{-\eta}, e^{\eta}\big], \qquad \eta = \frac{R\varepsilon}{h} + \frac{\varepsilon^2}{2h^2},$$

and for $\|u - U_i\| > Rh \ge 2\varepsilon$, $K_h(u - \hat{U}_i) \le (2\pi h^2)^{-k/2} e^{-\|u-U_i\|^2/8h^2}$. Since the density of U_τ is bounded below on D_τ and bounded above, and D_τ has Lipschitz boundary, the kernel mass of the near points is at least $c\,N$ and that of the far points at most $C\,N e^{-R^2/16}$, uniformly in u within $o(h)$ of D_τ, with probability tending to one (uniform laws of large numbers for kernel classes). Hence $\sup_u |\hat{\psi}(u) - \check{\psi}(u)| \le 2B\big(e^{2\eta} - 1 + C' e^{-R^2/16}\big)$, which tends to zero in probability when $R = R_N \to \infty$ slowly enough that $R\varepsilon/h \to 0$, possible by (A4).

Step 3: the test-time cue. At test time the tree is queried at $\hat{u}_\tau(x)$, so $|\hat{m}_\tau(x) - m_\tau(x)| \le \sup_{D_\tau^{(\varepsilon)}} |\hat{\psi} - \psi_\tau| + \omega_{\psi_\tau}(\varepsilon)$, where $D_\tau^{(\varepsilon)}$ is the ε-neighbourhood of D_τ; both terms tend to zero in probability by Steps 1–2 and (A2). This is the induction hypothesis for the children of τ; at the output tree it is the claim. □

Corollary 25 (Consistency if and only if sufficiency)**.** *Under the hypotheses of Theorem 24, $\mathbb{E}[(\hat{F}_N(X) - f(X))^2] \to \Delta(A)$. The forest is consistent for f if and only if its task assignment is sufficient.*

Proof. $\hat{F}_N$ and f are bounded by B, so uniform convergence in probability gives $\mathbb{E}(\hat{F}_N - F_A)^2 \to 0$, and $\mathbb{E}(\hat{F}_N - f)^2 \to \mathbb{E}(F_A - f)^2 = \Delta(A)$. □

Theorem 24 separates what Cobweb's learning does from what the tasks determine: every tree learns the conditional expectation of its outcome given its cue, nothing more and nothing less; whether the composition of these conditional expectations is f is a property of the task assignment alone.

5.3 Which tasks are sufficient

Funnel tasks. A convenient sufficient condition for funnel forests is that each tree's ideal message equals $\mathbb{E}[Y \mid X_{R(\tau)}]$, where $R(\tau)$ is the set of raw coordinates upstream of τ; this holds by induction when every $\mathbb{E}[Y \mid X_{R(\tau)}]$ is a function of the parents' $\mathbb{E}[Y \mid X_{R(\sigma)}]$ and of X_{S_τ}.

Proposition 26 (Funnel tasks are sufficient for monotone-additive targets)**.** *Let X consist of independent blocks $X_{B_1}, \dots, X_{B_m}$, and let $Y = \Phi\big(\sum_j g_j(X_{B_j})\big) + \xi$ with Φ continuous and strictly increasing, g_j continuous, and ξ centred and independent of X. Then every funnel activation forest whose graph is a rooted tree in which each block is the raw cue of exactly one tree is sufficient.*

Proof. For an upstream set of blocks S, $\mathbb{E}[Y \mid X_S] = \mathbb{E}[\Phi(s_S + Z_{\bar{S}})] =: H_S(s_S)$ with $s_S = \sum_{j \in S} g_j(X_{B_j})$ and $Z_{\bar{S}}$ the sum over the other blocks, independent of X_S; H_S is continuous and strictly increasing because Φ is. A tree whose children cover the disjoint block sets $S_1, \dots, S_k$ and which reads the raw blocks S_0 has $s_S = \sum_i H_{S_i}^{-1}(m_{S_i}) + \sum_{j \in S_0} g_j(X_{B_j})$, so its $\mathbb{E}[Y \mid X_S] = H_S(s_S)$ is a continuous function of its parents' messages and its raw cue. □

The KAN examples $\exp(\sin \pi x_1 + x_2^2)$, $\exp\big(\frac{1}{100}\sum_i \sin^2(\pi x_i/2)\big)$ and $\exp\big(\frac{1}{2}(\sin \pi(x_1^2 + x_2^2) + \sin \pi(x_3^2 + x_4^2))\big)$ (the last with the blocks (x_1, x_2) and (x_3, x_4)) are in this class.

Proposition 27 (Funnel tasks can fail)**.** *Let X_1, X_2 be independent and symmetric about 0, and $Y = X_1X_2$. For the funnel forest whose output tree reads the activations of a tree perceiving X_1 and a tree perceiving X_2, $m_1 = m_2 \equiv 0$, $F_A \equiv 0$ and $\Delta(A) = \mathrm{Var}(Y)$: the forest learns nothing, whatever N and whatever structure its trees build.*

Proof. $\mathbb{E}[X_1X_2 \mid X_1] = X_1\mathbb{E}X_2 = 0$, and likewise for X_2; the output tree's ideal cue is constant. By Theorem 18 the source trees' activations converge to 0. □

Cross-view tasks. A tree that predicts another part of the input never sees the target. It recovers what the views share, which suffices when the target depends on the input only through what the views share—the multi-view redundancy assumption of co-training and of contrastive learning [2, 22].

Proposition 28 (Cross-view tasks are sufficient under multi-view redundancy)**.** *Let $Z \in \mathbb{R}^r$ be latent and $X = (X^{(1)}, \dots, X^{(m)})$ consist of m views with (Z, X) jointly Gaussian and centred, $X^{(v)} = \Lambda_v Z + E_v$, the noises E_v independent of each other and of Z. Let $Y = g(Z) + \xi$ with $\mathbb{E}|g(Z)| < \infty$ and ξ centred and independent of (Z, X). Give the tree of view v the cue $X^{(v)}$ and as outcome r coordinates W_v of other views whose loading matrix $\Lambda_{W_v} \in \mathbb{R}^{r\times r}$ is invertible, and let the output tree have the activations of the m view trees as cue and Y as outcome. This assignment is sufficient, and no tree but the output tree perceives Y.*

Proof. Write $\mu_v = \mathbb{E}[Z \mid X^{(v)}]$ and P_v for the posterior precision given $X^{(v)}$, and P_0 for the prior precision; both precisions are constants. Since $W_v = \Lambda_{W_v} Z + E'$ with E' independent of $(Z, X^{(v)})$, the ideal message is $m_v = \mathbb{E}[W_v \mid X^{(v)}] = \Lambda_{W_v}\mu_v$, an invertible linear function of μ_v. By conditional independence of the views, $p(z \mid x) \propto p(z)\prod_v p(x^{(v)} \mid z) \propto p(z)^{1-m}\prod_v p(z \mid x^{(v)})$, a Gaussian with precision $P = \sum_v P_v - (m-1)P_0$ and mean $P^{-1}\sum_v P_v\mu_v$. The posterior of Z given X therefore depends on X only through $(\mu_v)_v$, hence through $(m_v)_v$, and so does $f(X) = \mathbb{E}[g(Z) \mid X]$. Thus $F_A = \mathbb{E}[Y \mid (m_v)_v] = f$. □

The assumption is necessary in the following sense: if the target carries a view-specific component, e.g. $Y = e^\top E_1$ with $X^{(1)}$ of dimension larger than r and e generic, the cross-view messages discard it and the deficit is positive. In images and text the views are patches and contexts, and the cross-view outcome is a neighbouring pixel or the next token.

Categorization. A tree without an outcome passes labels. Its labels do not depend on f at all, which is their strength and their cost.

Theorem 29 (Consistency of label forests)**.** *Let a forest consist of trees without outcomes over the raw coordinates passing labels at depth k_N (or leaf labels) to an output tree that models (labels, Y), learned with the split protocol, and let the output tree predict with the leaves-only LIK mixture with $\alpha_N \to 0$ fast enough ($\alpha_N \cdot \#\text{labels per tree} \cdot n \to 0$). Let $\mathcal{P}_N$ be the product partition of $[0,1]^n$ induced by the frozen label maps. If (a) the diameter of the cell of $\mathcal{P}_N$ containing X tends to 0 in probability and (b) the number of cells of $\mathcal{P}_N$ that contain a sample of the output chunk is $o(N)$, then the forest is a consistent regression estimate.*

Proof sketch. As in the proof of Theorem 15, with $\alpha_N \to 0$ the output tree's leaves-only mixture for a query in cell B puts weight $1 - o(1)$ uniformly on the leaves whose label tuple equals that

of B (one per training point of the output chunk in B, by Lemma 17); their y-means average to the partitioning estimate $\sum_{X_i \in B} Y_i / \#\{X_i \in B\}$. Conditional on the frozen partition this is the classical partitioning estimate, consistent under (a) and (b) [11, Thm. 4.2]. □

Remark 30 (The trade-off). Condition (b) requires the product partition to have fewer occupied cells than samples; with K labels per tree that is $K^n = o(N)$, so the rate of a label forest is governed by n, not by the fan-in. Funnel and cross-view forests have rates governed by the fan-in (Theorem 24 with Remark 14) but need sufficiency (Proposition 27).

Conjecture 31 (Hybrid messages)**.** *A forest whose trees pass labels and funnel activations is consistent for every continuous f (through its labels) and attains the fan-in rate when its funnel assignment is sufficient (through its activations).*

Remark 32 (When should a forest beat a single tree?). Comparing the upper bounds suggests where to look, without proving a separation. A noise-free target lets a single tree interpolate, with interior error of order $N^{-2/n}$ for a C^2 target; each tree of a funnel forest regresses an outcome that is noisy given its own cue (the coordinates it does not perceive act as noise, of variance $\mathrm{Var}(Y \mid X_{R(\tau)})$), with error of order $N^{-2/(k+4)}$. The bounds favour the forest only when $n > k + 4$. For a noisy target the single tree's bound is $N^{-2/(n+4)}$ and the comparison favours the forest as soon as $n > k$.

6 Inferring tasks

Corollary 25 reduces learnability to choosing a sufficient task assignment. The assignment is part of the forest's structure; we ask whether it can be inferred from the stream, and by what rule.

6.1 Tasks cannot be inferred locally

Definition 33 (Local task rule)**.** A *local task rule* decides something about a tree τ—keep or prune it, which task to give it, which message to pass—as a measurable function of τ's own training stream (the instances $(C_{\tau,i}, V_{\tau,i})$, $i \le N$, for a candidate outcome V_τ, possibly $\varnothing$) and of independent randomness. The tree's own prequential score, its category utility and its learned concepts are all functions of this stream.

Theorem 34 (Locality obstruction)**.** *Let X_1, X_2, X_3 be independent and uniform on $[-1, 1]$, observe only (X_1, X_2, Y), and consider the two worlds*

$$P_a : \ Y = \mathrm{sign}(X_1 X_2), \qquad P_b : \ Y = \mathrm{sign}(X_1 X_3).$$

(i) For $j \in \{1, 2\}$, the pair (X_j, Y) has the same law under P_a and P_b: X_j and Y are independent and Y is uniform on $\{-1, 1\}$. Hence, for every candidate outcome, the training stream of a tree with cue X_j has the same law in both worlds, and so does the output of every local task rule. (ii) Under P_a, $f(X) = Y$ and discarding X_2 loses everything ($\mathbb{E}\,\mathrm{Var}(Y \mid X_1) = 1$ while $\mathbb{E}\,\mathrm{Var}(Y \mid X) = 0$); under P_b, X_2 is irrelevant ($f(X) = \mathbb{E}[Y \mid X_1] = 0$). Consequently every local rule that decides whether to keep the tree perceiving X_2 errs with probability at least $1/2$ in one of the two worlds. (iii) Under P_a the funnel activations of both univariate trees vanish, so every funnel forest built on them has the maximal deficit $\Delta = \mathrm{Var}(Y) = 1$, whereas the label forest of Theorem 29 and a single tree over (X_1, X_2) are sufficient.

Proof. (i) Under P_a, $Y = \mathrm{sign}(X_1)\,\mathrm{sign}(X_2)$; given X_2, the factor $\mathrm{sign}(X_1)$ is uniform on $\{-1, 1\}$ and independent of X_2, so Y is uniform and independent of X_2; by symmetry the same holds for

X_1. Under P_b, $Y = \operatorname{sign}(X_1)\operatorname{sign}(X_3)$ is independent of X_2 outright and, by the same argument, of X_1. The stream of a tree with cue X_j is an i.i.d. sample of (X_j, V) with $V \in \{Y, \varnothing\}$ (or another observable whose joint law with X_j is the same in both worlds, such as X_{3-j}), so it has the same law in both worlds, and so does any measurable function of it. (ii) Direct. If p is the probability that the rule keeps the tree, it errs with probability $1-p$ under P_a and p under P_b. (iii) $\mathbb{E}[Y \mid X_j] = 0$ for $j = 1, 2$ by (i), so Proposition 27's argument applies; the label and single-tree statements follow from Theorems 18 and 29. □

Corollary 35 (Parity requires composition)**.** *Let $X_1, \dots, X_n$ be independent and uniform on $[-1, 1]$ and $Y = \prod_{j \in J} \operatorname{sign}(X_j)$. For every S that does not contain J, Y is independent of X_S. Hence no tree (and no rule using the streams of any set of trees) whose joint cue misses a coordinate of J can detect the dependence, every funnel activation of such a tree vanishes, and a forest learns Y only if some tree composes—by labels or by a merged cue—information from all of J.*

Proof. For $j \in J \setminus S$, $\operatorname{sign}(X_j)$ is uniform on $\{-1, 1\}$ and independent of X_S and of the other factors, so Y is uniform and independent of X_S. □

Theorem 34 is not a peculiarity of Cobweb: synergistic information—information about Y held by a set of variables jointly and by none of them alone [23]—is invisible to any module that sees one of them. Back-propagation escapes it because each module receives the error of the composition. A Cobweb tree receives no error, so the decision has to be taken where the information is composed.

6.2 Verification at the point of composition

Theorem 36 (Task selection by verification)**.** *Let $\mathcal{A}$ be a finite set of task assignments. For each $A \in \mathcal{A}$ let $\hat{F}_{A,N}$ be the output of the forest that Cobweb learns with assignment A from a training stream of length N, with $|\hat{F}_{A,N}| \le B$ and $\mathbb{E} \int |\hat{F}_{A,N} - F_A|^2 d\mu \to 0$ (Theorems 18, 24 and 29 give this for activation forests, label forests and single trees; for a single tree, and for a label forest under the conditions of Theorem 29, $F_A = f$). Let $\hat{A}_N$ minimise the mean squared error of Y on an independent validation stream of length $N_v(N) \to \infty$. Then*

$$\Pr\Big[\Delta(\hat{A}_N) = \min_{A \in \mathcal{A}} \Delta(A)\Big] \to 1.$$

In particular, if $\mathcal{A}$ contains a sufficient assignment, the selected assignment is sufficient with probability tending to one and the selected forest is consistent.

Proof. Let $\sigma^2 = \mathbb{E}\operatorname{Var}(Y \mid X)$. Conditionally on the training stream, the expected validation error of A is $R_N(A) = \sigma^2 + \mathbb{E}[(f - \hat{F}_{A,N})^2 \mid \text{train}]$, and $\mathbb{E}[(f - \hat{F}_{A,N})^2 \mid \text{train}] \to \Delta(A)$ in probability, because $\|f - \hat{F}_{A,N}\|_{L_2(\mu)} - \|f - F_A\|_{L_2(\mu)}$ is bounded by $\|\hat{F}_{A,N} - F_A\|_{L_2(\mu)} \to 0$. The validation average of the bounded squared errors differs from $R_N(A)$ by $O_P(N_v^{-1/2})$ (Chebyshev). With $\gamma = \min\{\Delta(A) - \Delta^* : \Delta(A) > \Delta^*\} > 0$, $\Delta^* = \min_A \Delta(A)$, every assignment with $\Delta(A) > \Delta^*$ has a larger validation error than every minimiser with probability tending to one, and $\mathcal{A}$ is finite. □

The validation error at the output is the outcome of the composing tree. With the log loss in place of the squared error it estimates the conditional entropy $H(Y \mid U_o)$, so Theorem 36 selects the task assignment that maximises the predictive information $I(Y; U_o)$—the same quantity, mutual information, that category utility maximises when Cobweb chooses among its operators, evaluated one level up. In the online protocol the prequential error $\frac{1}{N}\sum_i (Y_i - \hat{F}_{A,i-1}(X_i))^2$ plays the role of the validation error [5]; it converges to the same limit whenever the online forest's risk tends to that of F_A, by a Cesàro argument.

6.3 Task operators

Definition 37 (Task operators and synergy deficit)**.** Let a tree with outcome V compose two subtrees σ_1, σ_2 with cues C_1, C_2. The *task operators* at this composition are

- ACT: σ_j gets the outcome V and passes its activation;
- LABEL: σ_j gets no outcome and passes its label;
- HYBRID: σ_j gets the outcome V and passes its activation and label;
- CROSS: σ_j gets a coordinate of C_{3-j} as outcome and passes its activation;
- MERGE: σ_1, σ_2 are replaced by one tree with cue $C_1 \cup C_2$.

The *synergy deficit* of the pair is $\Delta_{\text{ACT}}(\sigma_1, \sigma_2) = \mathbb{E}\,\mathrm{Var}(V \mid a_1, a_2) - \mathbb{E}\,\mathrm{Var}(V \mid C_1, C_2) \geq 0$, with $a_j = \mathbb{E}[V \mid C_j]$; it is zero exactly when ACT loses nothing at this composition, and it is estimated by the difference of the validation errors of ACT and MERGE.

MERGE and its inverse mirror Cobweb's own merge and split one level up: they decide whether two cues are perceived by one concept hierarchy or by two.

Procedure 38 (Task induction by verification)**.**

1. Start with one tree per raw coordinate (or per given block).
2. Estimate the synergy deficit of every pair; merge the pairs whose deficit is significantly positive (union–find over these pairs).
3. Compose the resulting groups bottom-up with two-input trees; at each composition choose the operator in $\{\text{ACT}, \text{LABEL}, \text{HYBRID}, \text{CROSS}\}$ by the validation error of the composing tree's outcome.

By Theorem 36 each choice in step 3 is consistent given the subtrees below it; the procedure as a whole is greedy and carries no global guarantee. By Theorem 34 no step can be replaced by a rule that looks at a subtree alone. Every tree still learns only with `ifit`: the procedure chooses *what* a tree predicts—which observable it will see—never how it learns.

7 Experiments

The experiments test the statements of Sections 3 to 6 and repeat the KAN paper's battery [13]. They do not claim that Cobweb forests are competitive regressors: a Cobweb tree is a nonparametric smoother, and on smooth low-dimensional targets the gradient-trained MLP and KAN are far more accurate.

Setup. Trees are the hybrid Cobweb of `cobweb-private` with the additions listed in the repository (masked queries, node identifiers, leaf-counted budgets); a test checks that it reproduces the original trees and predictions exactly. Unless stated otherwise every tree predicts with the leaves-only PATH mixture and a budget of 300 leaves, every forest is trained with the online protocol (Definition 6), continuous attributes are measured in units of their training standard deviation, and the variance floor is $h^2 = (\kappa N^{-1/(d+4)})^2$ for a tree with d continuous cues, with κ (and the label depth) chosen on a fresh validation set. MLP and KAN (B-spline edges as in Liu et al. [13]) are trained with Adam and early stopping, with width or grid chosen on the same validation set. Toy targets are noise-free on $[-1, 1]^n$ and are scored on 2000 fresh points; errors are RMSE divided by the standard deviation of f, averaged over three seeds (five for Section 7.3). Every result file records the hashes of the code that produced it, and the verifier checks that every reported result comes from a build without the budget bug described in the repository (Appendix A).

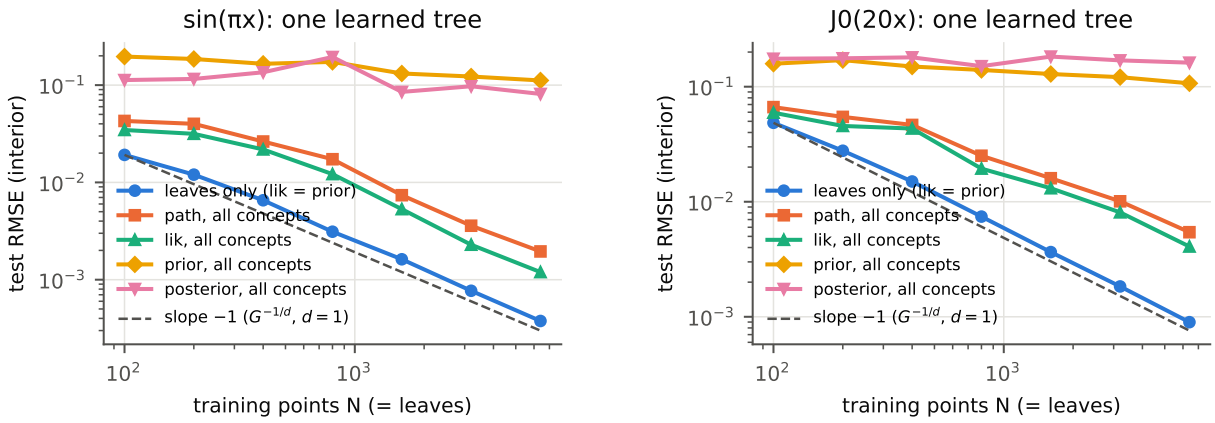


Figure 1: One learned tree (variance floor at the data spacing, full expansion). The leaves-only mixture follows the slope -1 of (3) for $d = 1$; the PATH and LIK mixtures over all concepts converge more slowly; the PRIOR and POSTERIOR mixtures over all concepts plateau (Proposition 10).

7.1 One tree

Figure 1 tests Theorems 9 and 18 and Proposition 10 on learned trees. For $\sin(\pi x)$ the leaves-only error falls from 0.019 at $N = 100$ to 0.0004 at $N = 6400$, a factor of two per doubling, while the PRIOR mixture over all concepts stays between 0.11 and 0.20 and the POSTERIOR mixture between 0.08 and 0.19. $J_0(20x)$ behaves the same way.

7.2 The KAN battery

Figure 2 and Table 1 show four things. (i) One tree is consistent on every target of dimension at most four, as Theorem 18 predicts, but slowly: on $\sin(\pi x)$ its error at $N = 16000$ (0.003) is several times the MLP's and KAN's. (ii) On xy the funnel forest stays at 1.00 for every N, exactly as Proposition 27 predicts, while labels (0.22 at $N = 16000$) and hybrid messages (0.19) learn; the single tree over both coordinates is better still (0.011), because with $n = 2$ it has the same fan-in. (iii) On the 4-D target the funnel forest over the pairs $(x_1, x_2), (x_3, x_4)$, which is sufficient by Proposition 26, beats the single tree at small N (0.46 against 0.57 at $N = 1000$) and loses at $N = 16000$ (0.33 against 0.28), the crossover that Remark 32 suggests for a noise-free target; with a noisy target the two tie at $N = 16000$. (iv) On the 100-D target every Cobweb design stays near 1 while KAN reaches 0.002.

Table 2 locates the 100-D failure. Each univariate tree must estimate $\mathbb{E}[Y \mid X_i]$, whose standard deviation is about $1/7$ of that of Y. With a budget of 100 leaves the estimate is dominated by noise (ratio 1.4); with 1000 or 4000 leaves it carries signal (ratio 0.57), but the trees one level up read these noisy activations, and by the third level the signal is gone. This is assumption (A4) of Theorem 24 failing at finite N: the parents' errors are not small compared with the children's bandwidth, and errors compound through the levels. A KAN does not face this problem, because its gradient corrects all inner functions against the error of the output.

7.3 Cognitive tasks

Locality. Table 3 reproduces Theorem 34 in data: the tree that perceives x_2 has the same validation error and the same flat activation in both worlds, so nothing it can compute tells it

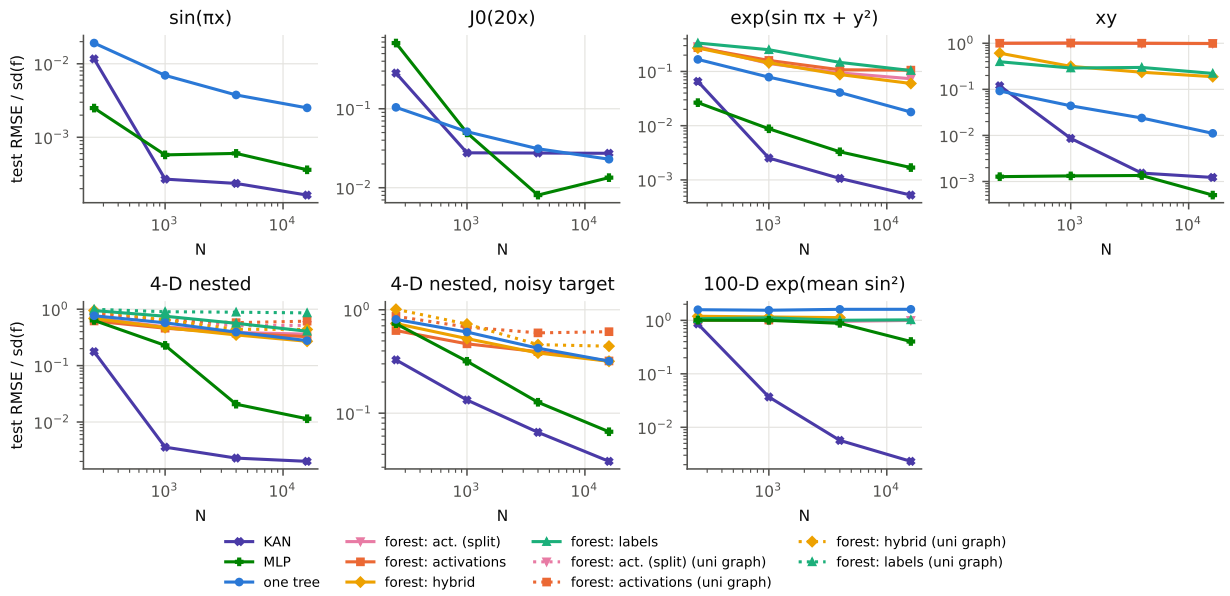


Figure 2: Test RMSE/sd(f) against N. Forests are drawn with their best graph (dotted: the univariate graph of the 4-D target). The activation forest of xy stays at 1 (Proposition 27).

whether x_2 matters. The composing tree separates the worlds: under P_a the merged tree reaches 0.22 and the label forest 0.43, while under P_b every assignment stays near 1. The funnel forest fails in both worlds.

Selection. Selecting by the validation error at the output (Table 4) never selects the funnel forest on xy or on the sign target, and on the 4-D target it selects the sufficient pairs forest over every two-level univariate assignment and over the merged 4-D tree. In every cell the selected candidate is the one with the lowest test error or within one percent of it. On two-dimensional targets the merged tree wins, as it should: it has the same fan-in as the forest and no composition error.

Synergy. Figure 3 estimates the synergy deficit of every pair of coordinates, the quantity step 2 of Procedure 38 thresholds. On the 4-D target the within-block deficits are 0.18 and 0.23 of $\mathrm{Var}(Y)$ and the cross-block deficits are within 0.03 of zero, so the procedure recovers the blocks $\{x_1, x_2\}$ and $\{x_3, x_4\}$. On housing the largest deficits are longitude with latitude (0.14 at $N = 4000$) and rooms with bedrooms (0.13): location and dwelling size are the interactions a funnel forest cannot represent through activations, which is why the pairs graph wins on housing below. At $N = 16000$ every housing pair shows a positive deficit (between 0.03 and 0.22), so the threshold of step 2 must account for sample size.

Cross-view tasks. In the two-view model (Fig. 4) the cross-view forest, whose inner trees never see Y, keeps improving when the target depends on the shared latent (0.45 at $N = 500$, 0.27 at $N = 32000$), as Proposition 28 predicts, while the funnel forest stalls near 0.39: here the funnel messages $\mathbb{E}[\sin 2Z \mid X^{(v)}]$ are not invertible functions of the per-view posterior means, so the funnel assignment is not sufficient. When the target is specific to one view the cross-view forest stalls near 0.75, the deficit the remark after Proposition 28 predicts. The single tree over all four coordinates is best in both panels at these sample sizes.

fn	N	one tree	activations	labels	hybrid	act. (split)	MLP	KAN
sin	250	0.019	–	–	–	–	0.002	0.012
sin	1000	0.007	–	–	–	–	0.001	0.000
sin	4000	0.004	–	–	–	–	0.001	0.000
sin	16000	0.003	–	–	–	–	0.000	0.000
j0	250	0.104	–	–	–	–	0.679	0.283
j0	1000	0.051	–	–	–	–	0.050	0.028
j0	4000	0.031	–	–	–	–	0.008	0.028
j0	16000	0.023	–	–	–	–	0.013	0.027
expsin	250	0.167	0.274	0.334	0.271	0.289	0.027	0.065
expsin	1000	0.079	0.160	0.252	0.145	0.140	0.009	0.003
expsin	4000	0.041	0.107	0.147	0.088	0.094	0.003	0.001
expsin	16000	0.018	0.106	0.104	0.060	0.073	0.002	0.001
xy	250	0.092	1.002	0.397	0.610	0.997	0.001	0.119
xy	1000	0.044	1.011	0.291	0.315	1.004	0.001	0.009
xy	4000	0.024	1.003	0.299	0.235	0.991	0.001	0.002
xy	16000	0.011	0.991	0.222	0.188	0.985	0.001	0.001
nested4	250	0.765	0.618	0.946	0.685	0.695	0.633	0.176
nested4	1000	0.574	0.460	0.751	0.472	0.488	0.228	0.004
nested4	4000	0.393	0.376	0.563	0.350	0.386	0.021	0.002
nested4	16000	0.277	0.326	0.407	0.270	0.361	0.011	0.002
nested4 (noisy)	250	0.807	0.630	–	0.735	–	0.733	0.328
nested4 (noisy)	1000	0.609	0.468	–	0.526	–	0.317	0.134
nested4 (noisy)	4000	0.425	0.393	–	0.382	–	0.128	0.065
nested4 (noisy)	16000	0.319	0.320	–	0.318	–	0.066	0.034
exp100	250	1.580	1.007	1.073	1.192	1.027	1.007	0.856
exp100	1000	1.546	1.015	1.114	1.156	1.004	1.000	0.037
exp100	4000	1.606	0.992	1.003	1.125	0.957	0.880	0.006
exp100	16000	1.609	–	1.022	–	0.996	0.402	0.002

Table 1: The KAN battery: test RMSE/sd(f), mean of three seeds. "activations" is the funnel forest; for the 4-D target the forests use the pairs graph (superscript u: univariate graph).

7.4 Continual learning, interpretability and labels

No forgetting. Lemma 17 says that nothing a tree has learned is deleted. On the KAN paper's continual task (Fig. 5 and Table 5) the tree's error on everything seen so far stays between 0.037 and 0.048 through all five phases; the MLP forgets (its error reaches 10.8), and the KAN with a fine grid forgets partly (0.17).

Interpretability. As a KAN's edges can be plotted, so can a Cobweb forest's trees. On $\exp(\sin \pi x_1 + x_2^2)$ the univariate trees learn $c_1 e^{\sin \pi x_1}$ and $c_2 e^{x_2^2}$ (Fig. 6), the inner functions of the target up to the monotone maps H_S of Proposition 26, without gradients.

Label cells. Theorem 29 needs the cells of the labels to shrink. In one dimension the cells of a fixed depth shrink slowly as the tree grows; in two dimensions they do not (Fig. 7). Labels therefore become fine enough only if their depth grows with N, which the selection experiments

N	leaf budget	κ	level 0 $\lvert S\rvert\leq 1$	level 1 $\lvert S\rvert\leq 2$	level 2 $\lvert S\rvert\leq 4$	level 3 $\lvert S\rvert\leq 8$	level 4 $\lvert S\rvert\leq 16$	level 5 $\lvert S\rvert\leq 32$	level 6 $\lvert S\rvert\leq 64$	output RMSE/sd
4000	100	0.25	1.38	1.42	1.40	1.27	1.12	1.08	1.02	1.00
4000	100	1.0	3.16	1.46	1.41	1.38	1.19	1.08	1.06	1.01
4000	1000	1.0	0.57	0.87	1.06	1.10	1.06	1.06	1.02	1.00
4000	4000	1.0	0.57	0.82	0.99	1.02	1.01	1.00	1.00	1.00

Table 2: The 100-D funnel forest level by level (N training points, one seed): RMSE of every tree's activation divided by the standard deviation of its ideal message m_S, averaged over the trees of a level. A ratio near 1 means the activation carries no information about m_S.

		tree perceiving x_2 alone		composing tree (test RMSE)			
N	world	val. RMSE	act. sd	funnel	labels (d=3)	hybrid	merged
1000	P_a	1.004	0.062	1.008	0.675	0.642	0.270
1000	P_b	1.002	0.080	1.006	1.143	1.116	1.080
4000	P_a	1.000	0.047	1.064	0.460	0.510	0.244
4000	P_b	1.002	0.048	1.028	1.074	1.056	1.018
16000	P_a	1.003	0.069	1.164	0.429	0.471	0.215
16000	P_b	1.004	0.073	1.092	1.051	1.034	1.007

Table 3: The two sign worlds of Theorem 34 (five seeds). Left: a tree that perceives x_2 and predicts Y—its validation RMSE equals that of the constant predictor (1.00) and its activation is flat in both worlds. Right: the composing tree's test RMSE for four task assignments.

do by choosing the depth on validation data. We also tried the cognitively attractive choice of the *basic level*—the concept of largest expected pointwise mutual information on the branch, which Cobweb computes: for continuous attributes it lies near the leaves (mean depth 10 to 11), because a concept's information grows as its variance shrinks towards the variance floor, and its labels did worse than any fixed depth on xy.

7.5 Real data

Feynman equations. With a balanced graph over the inputs in their listed order—a graph that ignores the structure of each formula—the forests lose to the single tree on every equation (Table 6), and both lose to the MLP and the KAN by one to two orders of magnitude. A forest needs a sufficient graph; Procedure 38 is the way to find one, and we have not yet run it on these equations.

Housing. The funnel forest over the natural pairs (longitude–latitude, age–rooms, bedrooms–occupancy, households–income) beats the single tree at every N (0.276 against 0.303 at $N = 16000$; Table 7), and the univariate graph does not, consistent with the synergy deficits of Fig. 3. MLP and KAN are better still.

MNIST. Patch trees that predict the class and pass their predicted class distributions beat the single tree over all pixels at every N, and the hybrid forest reaches 94.9% at $N = 20000$, level with the KAN (94.5%) and below the MLP (96.8%; Table 8). Unsupervised patch labels (the Convolutional-Cobweb protocol) are the weakest design at every N: categorization alone passes no information about the class.

target	N	selected (count of 5)	funnel	selected
xy	500	merge (5)	1.020	0.071
xy	2000	merge (5)	1.000	0.032
xy	8000	merge (5)	1.003	0.019
sign	500	merge (5)	0.996	0.309
sign	2000	merge (5)	1.007	0.245
sign	8000	merge (5)	1.075	0.231
expsin	500	merge (5)	0.232	0.115
expsin	2000	merge (5)	0.145	0.055
expsin	8000	merge (5)	0.111	0.034
nested4	500	funnel-pairs (4), funnel (1)	0.773	0.662
nested4	2000	funnel-pairs (5)	0.651	0.451
nested4	8000	funnel-pairs (3), hybrid-pairs (2)	0.622	0.334

Table 4: Task selection by verification (Theorem 36; five seeds). The candidates are the funnel forest, label forests of depth 2, 3 and 4, a hybrid forest, one merged tree and, for the 4-D target, funnel and hybrid forests over the pairs. Columns: how often each candidate was selected, and the test RMSE/sd of the funnel forest and of the selected forest.

method	RMSE on all data seen so far, after phase 1..5				
cobweb	0.037	0.048	0.045	0.045	0.045
mlp	0.014	0.965	10.756	7.506	8.241
kan	0.009	0.107	0.161	0.160	0.168

Table 5: Continual learning: RMSE on all data seen so far after each phase (three seeds).

Text. On next-word prediction (Table 9) the trees predict the most frequent token and their cross-entropy is worse than that of an interpolated n-gram; the forests do not help at $N = 20000$. A Cobweb language model needs the architecture of a dedicated study; we report these numbers as the limit of the generic forest.

8 Discussion

A cognitive reading. The three task families of Definition 20 are three things a learner can do with what it perceives: *categorize* it, *predict an outcome* of it, or *predict another part of it.* The first is concept formation in Cobweb's original sense; the second is cue–outcome learning [19]; the third is the perceptual prediction of predictive-processing accounts [3]. Each has a mathematical counterpart here. Categorization is always sufficient but pays for every cell it distinguishes (Theorem 29 and Remark 30), and it becomes fine enough only if acuity sharpens with experience—the variance floor $h_N \to 0$ of every consistency result, which is perceptual learning in the sense of Goldstone [10]. Outcome prediction compresses the cue to what matters for the outcome, which is sufficient exactly when the outcome's dependence on the parts is monotone-additive (Proposition 26) and fails on interactions (Proposition 27). Perceptual prediction never needs the target and is sufficient when the target depends on what the views share (Proposition 28).

Credit assignment. Theorem 34 is a precise form of the credit-assignment problem [18]: a module cannot learn its role from the statistics it sees when the information it carries is synergistic. Back-propagation solves the problem by sending the composed error to every module. A Cobweb

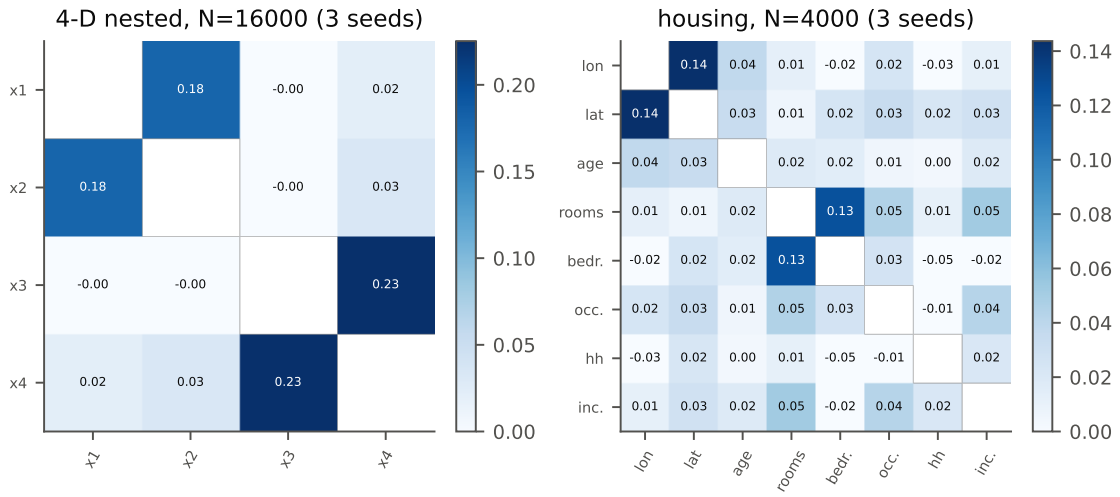


Figure 3: Synergy deficits (Definition 37): validation MSE of the funnel pair forest minus that of one tree over the pair, divided by $\mathrm{Var}(Y)$. Left: the 4-D target; the two blocks stand out and the cross-block deficits are near zero. Right: California housing; the largest deficits are longitude–latitude and rooms–bedrooms.

forest has no error signal, and the theory says where the decision must be taken instead: at the tree that composes the parts, by verifying candidate tasks against an observed outcome (Theorem 36). The locality and selection experiments show both halves.

Limitations. Theorem 24 covers the split protocol with full expansion and the LIK mixture; the experiments use the online protocol, a leaf budget and the PATH mixture, which the theory does not cover (Remark 19). Theorem 29 is proved in outline only, and the hybrid case is a conjecture (Conjecture 31). Procedure 38 is greedy and its thresholds are not calibrated. Finally, the rates are those of nonparametric smoothing: on smooth targets gradient-trained models are far more accurate, and in high dimension a funnel forest's errors compound through its levels (Table 2).

A Verification

Executable verifier. `verification/checks.py` instantiates the objects each statement talks about with the real C++ tree—learned by `ifit`, or built exactly for the constructions in the proofs (depth-one grid trees, the label-forest construction, interwoven trees)—and tests the statement's conclusion; `verification/run.py` writes the table below. The first check compares Definition 3 and Definition 1, implemented from the definitions, with the code's scores and mixtures.

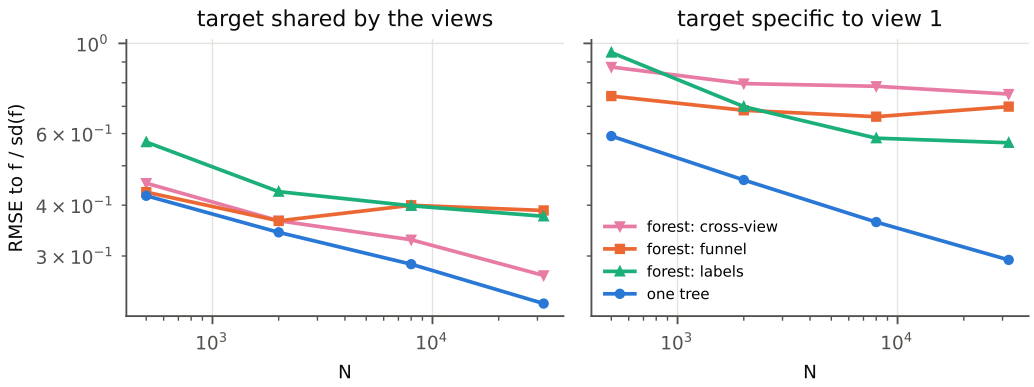


Figure 4: The two-view Gaussian model of Proposition 28 (five seeds): RMSE to the optimal predictor f, divided by $\mathrm{sd}(f)$. Left: $Y = \sin(2Z)$ depends on the shared latent. Right: Y is the first noise coordinate of view 1.

result	statement	check
pass	Definition 3	Def. multi-node prediction and Def. concept equal the implementation
pass	Lemma 17	Leaf preservation (no leaf deleted; additivity; leaves hold identical copies)
pass	Lemma 8	Concept-bias certificate holds for every weighting and budget
pass	Theorem 9, Proposition 10	Grid tree: error and root weight vanish (lik, path, leaves); prior plateaus
pass	Theorem 15	Label-forest construction meets its error bound; labels are nearest grid points
pass	Proposition 16	Interwoven tree = convex combination of the subtrees (lik, path)
pass	Lemma 11, Theorem 12	Composed constructed trees approximate $\Phi(\varphi_1 + \varphi_2)$
pass	Theorem 18	Learned tree, leaves-only lik = Nadaraya–Watson (path is not)
pass	Proposition 27	xy: funnel activations vanish, funnel forest fails; label error falls with N at every depth
pass	Theorem 34	Sign construction: local laws equal; composition separates the worlds
pass	Proposition 28	Posterior given all views = function of per-view posterior means
pass	Remark 19	Branch-and-bound stop returns a sub-optimal concept
pass	Proposition 26	H_S strictly increasing; funnel forest on a monotone-additive target converges
pass	Corollary 35	3-way parity: pairs independent of Y; label forest learns it, funnel does not
pass	Theorem 24	Split protocol: sup error to the ideal forest falls with N
pass	Theorem 36	Validation at the output never selects the funnel forest on xy

Adversarial review. Each statement is also checked by an independent reviewer instructed to break it; the reviewer is calibrated beforehand on a copy of this document with planted errors. The protocol and its results are in `verification/REVIEW.md`.

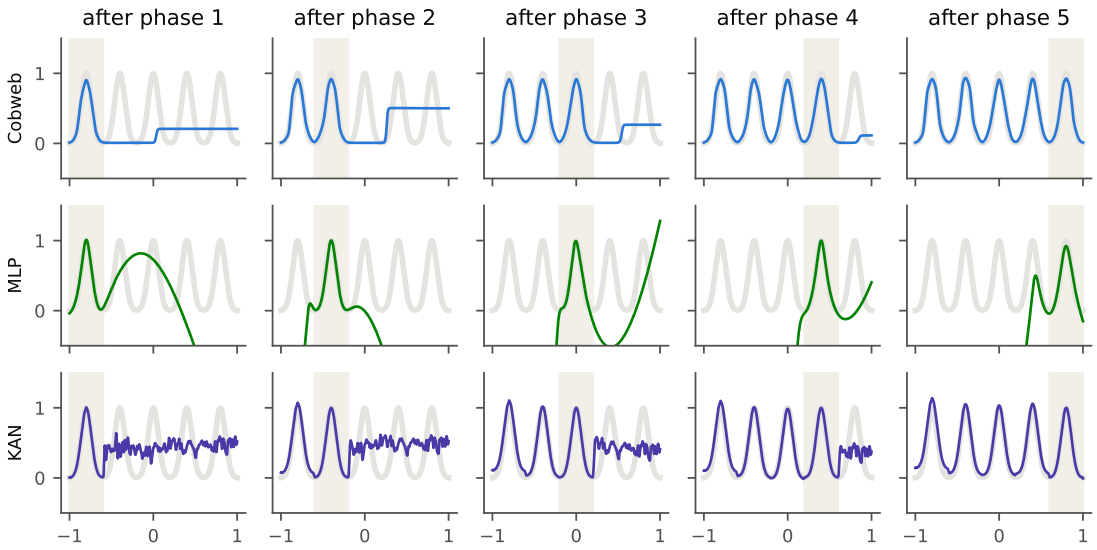


Figure 5: The continual-learning task of Liu et al. [13, Fig. 3.4]: five peaks, one region per phase (shaded). Grey: target.

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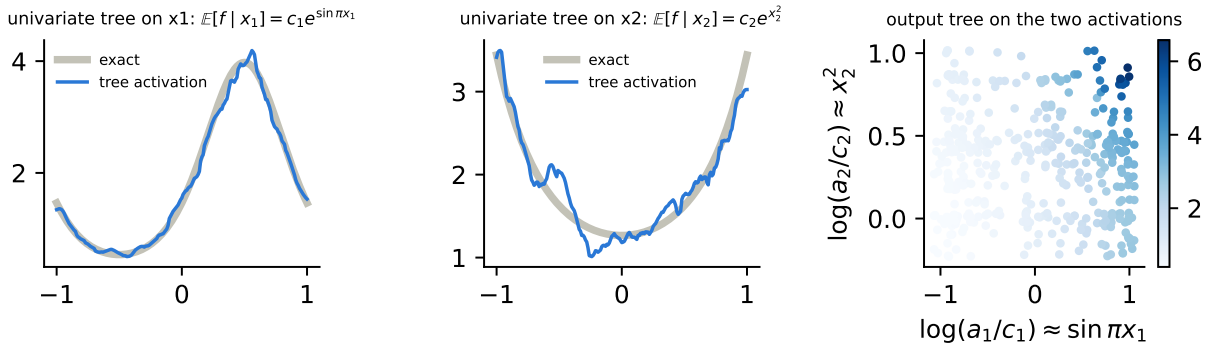


Figure 6: What the univariate trees of a funnel forest learn on $\exp(\sin \pi x_1 + x_2^2)$ ($N = 4000$). Left and centre: each tree's activation against the exact conditional expectation $\mathbb{E}[f \mid x_i]$. Right: the output tree's prediction as a function of the two activations.

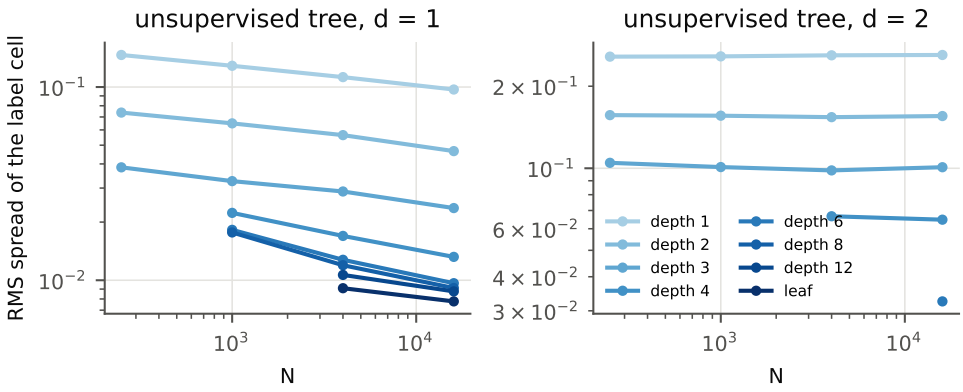


Figure 7: Spread of the cell of a label at fixed depth, for unsupervised trees over uniform data.

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equation	n	single	act	label	hybrid	mlp	kan
I.12.11	5	0.140	0.592	0.578	0.479	0.004	**0.002**
I.13.12	5	0.281	0.626	0.916	0.492	0.006	**0.005**
I.29.16	4	0.182	0.854	0.880	0.422	**0.011**	0.095
I.37.4	3	0.092	0.257	0.360	0.206	**0.003**	0.003
I.50.26	4	0.308	0.801	0.871	0.793	**0.016**	0.125
I.6.20	2	0.021	0.339	0.097	0.094	0.001	**0.001**
I.6.20a	1	0.002	–	–	–	0.000	**0.000**
I.6.20b	3	0.078	0.511	0.605	0.541	**0.003**	0.020
I.9.18	9	0.322	0.476	0.891	0.573	**0.003**	0.015
II.6.15a	6	0.486	0.556	0.878	0.421	**0.013**	0.024

Table 6: Feynman equations, $N = 4000$: test RMSE/sd(f), mean of three seeds; forests use a balanced graph over the inputs in their listed order.

N	single	act:pairs	act:uni	label:uni	label:pairs	hybrid:pairs	hybrid:uni	mlp	kan
1000	0.362±0.009	0.336±0.011	0.366±0.002	0.535±0.009	0.464±0.027	0.357±0.011	0.408±0.016	0.308±0.009	0.286±0.008
4000	0.325±0.006	0.299±0.005	0.344±0.005	0.496±0.035	0.461±0.005	0.314±0.008	0.339±0.012	0.271±0.005	0.262±0.004
16000	0.303±0.002	0.276±0.003	0.346±0.006	0.496±0.014	0.432±0.008	–	–	–	–

Table 7: California housing: test RMSE of the log median value (three seeds).

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N	single	label	act	hybrid	mlp	kan
1000	83.2±1.1	70.7±1.7	85.8±0.3	84.2±0.7	88.2±1.1	88.6±0.6
5000	91.4±0.4	81.9±1.2	92.4±0.2	92.3±0.5	94.0±0.2	92.2±0.7
20000	94.2±0.3	86.2±1.1	94.4±0.5	94.9±0.6	96.8±0.3	94.5±0.7

Table 8: MNIST, accuracy (%, three seeds). Forests have one tree per 7×7 patch; "label" is the Convolutional-Cobweb protocol, "act" passes each patch tree's predicted class distribution.

N	ngram	mlp	single	label-unsup	label-funnel
20000	17.8 / 5.07	20.9 / 5.10	20.9 / 5.88	20.9 / 5.87	20.9 / 5.77
100000	19.1 / 4.84	21.7 / 4.66	20.9 / 5.14	–	–
300000	19.9 / 4.68	23.6 / 4.41	–	–	–

Table 9: Next word on OpenWebText from the previous four words (2000-word vocabulary): top-1 accuracy (%) / cross-entropy (nats). An accuracy of 20.9 is the frequency of the unknown-word token.

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